

Project Report: Movie Rating Prediction

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1. Project Overview

The objective of this project was to create a data science solution capable of estimating the rating of a movie based on its metadata. By leveraging historical IMDb data for Indian cinema, the model identifies patterns between creative contributors (directors, actors), genres, and the final reception by the audience.

2. Dataset Analysis

The dataset "**IMDb Movies India.csv**" consists of several thousand entries representing the history of Indian films.

- **Key Features:** Movie Name, Release Year, Duration, Genre, Director, and lead Actors (1, 2, and 3).
- **Target Variable:** Rating (A numerical scale representing user/critic scores).
- **Challenges:** The raw data contained significant noise, including non-numeric characters in numerical columns and a high number of unique categorical values.
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3. Data Preprocessing & Cleaning

To prepare the data for machine learning, the following steps were performed:

- **Handling Missing Values:** Rows without a Rating (the target) were removed. Missing values in Duration and Votes were imputed using the median to maintain data integrity without introducing outliers.
- **String Formatting:**
 - The Year column was stripped of parentheses (e.g., "(2022)" → 2022).
 - Duration was converted to a float by removing " min".

- Votes were cleaned of commas and converted to numeric format.
- **Categorical Refinement:** The Genre column was simplified by extracting the primary genre listed for each film.

4. Feature Engineering

The most critical phase of the project involved **Target Encoding** for high-cardinality categorical features:

- **The Problem:** There are thousands of unique directors and actors. Using "One-Hot Encoding" would create a massive, sparse matrix that crashes standard models.
- **The Solution:** Each Director and Actor was replaced with their **historical average rating**. This transforms a name into a "reputation score," which the regression model can process effectively.

5. Machine Learning Modeling

Two distinct regression techniques were implemented to compare performance:

Model	Purpose	Strength
Linear Regression	Baseline	Provides a simple, interpretable relationship between features and ratings.
Random Forest Regressor	Final Model	Captures complex, non-linear interactions (e.g., a specific director working in a specific genre).

Mathematical Evaluation

The models were evaluated using three primary metrics:

1. **Mean Absolute Error (MAE):** The average magnitude of errors in the predictions.
2. **Mean Squared Error (MSE):** Penalizes larger errors more heavily.
3. **R-squared (R^2):** Measures how much of the variance in ratings is explained by our features.

6. Key Findings & Results

- **Feature Importance:** Analysis revealed that **Votes** and the **Director's historical rating** were the most significant predictors of a movie's final rating.
- **Model Performance:** The Random Forest Regressor consistently outperformed Linear Regression, suggesting that the relationship between movie features and success is non-linear and depends on the synergy of the cast and crew.

7. Conclusion

This project successfully demonstrates the power of machine learning in the entertainment industry. By cleaning historical data and applying advanced encoding techniques, we developed a model that provides a data-driven estimate of a movie's potential success.